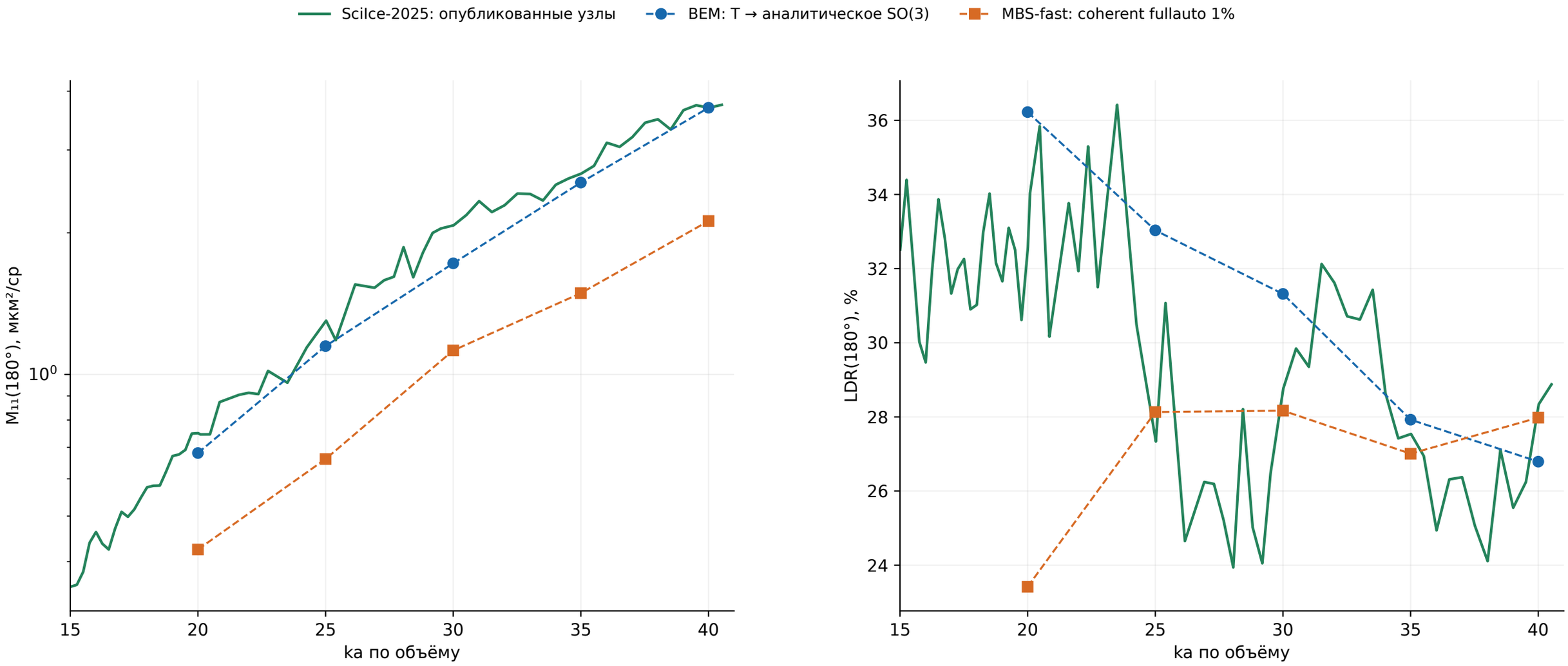
ScIce + BEM + MBS-fast: обратное рассеяние столбиков

m = 1,3116; λ = 0,532 мкм; D по вершинам/H = 0,7; случайная ориентация

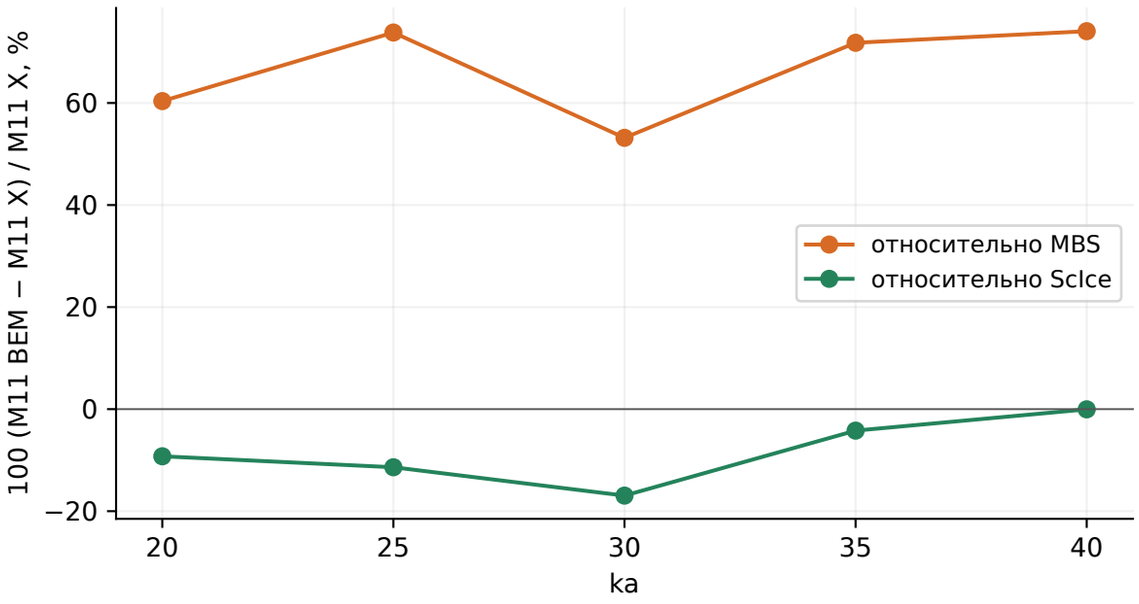


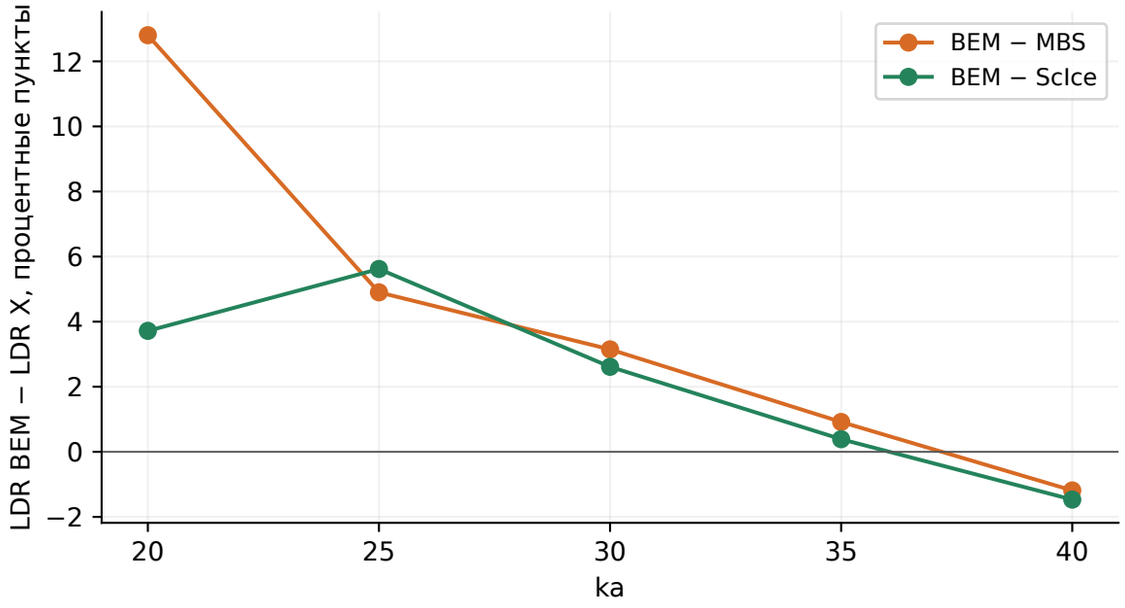
Пять BEM-решений рассчитаны напрямую. Масштаб не подгонялся; геометрия совпадает.
Штриховые отрезки соединяют узлы; они не являются проверенной интерполяцией.

ScIce: Konoshonkin et al. (2025), doi:10.5281/zenodo.15144868. Опубликованная таблица сохранена.

Расхождения при 180° на совпадающих размерах

SciIce: линейная интерполяция M11 и M22 к ka BEM; BEM и MBS: готовые значения в узлах.



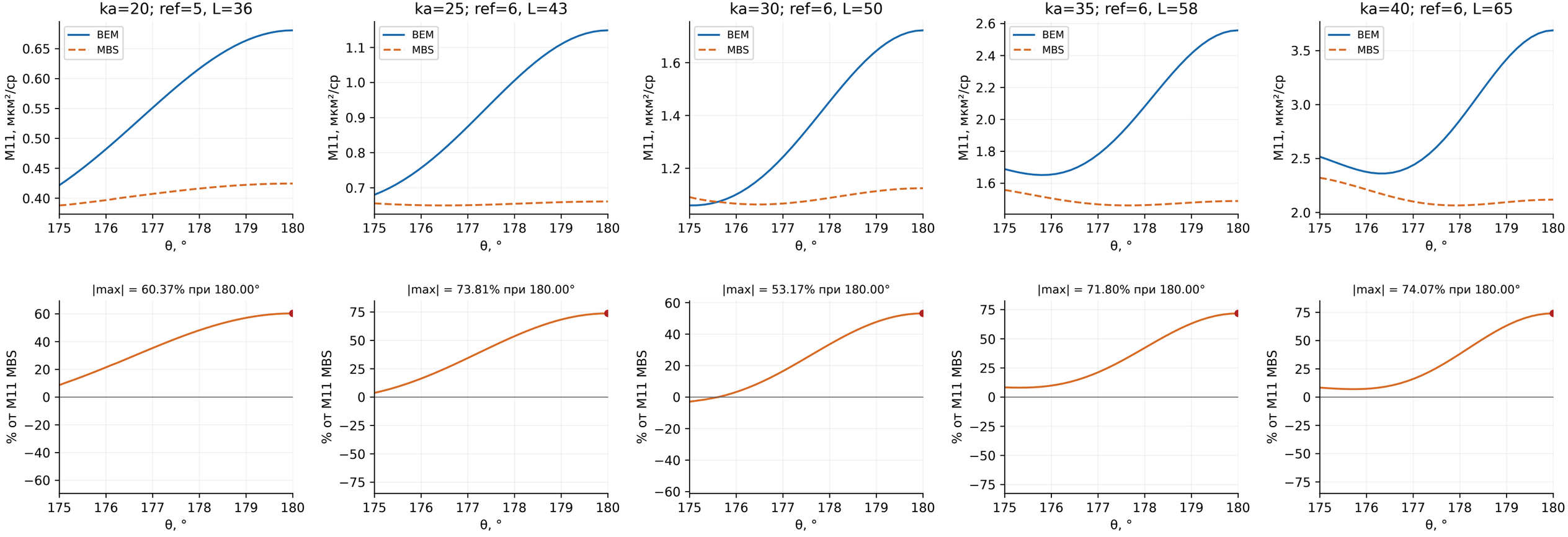


ka	M11 BEM	M11 MBS	M11 SciIce*	Δ/MBS	Δ/SciIce*
20	0.68078	0.42451	0.75010	+60.37%	-9.24%
25	1.14884	0.66098	1.29651	+73.81%	-11.39%
30	1.72246	1.12453	2.07391	+53.17%	-16.95%
35	2.55795	1.48891	2.67042	+71.80%	-4.21%
40	3.68870	2.11911	3.69024	+74.07%	-0.04%

BEM и MBS-fast: конус 175–180°, допуск fullauto 1%

m = 1,3116; λ = 0,532 мкм; D по вершинам/H = 0,7; случайная ориентация

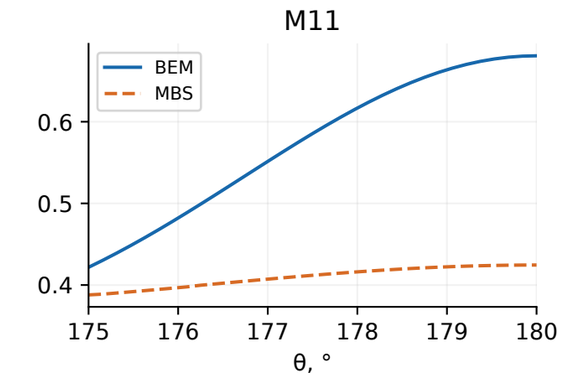
Ошибка: 100 (M11 BEM − M11 MBS) / M11 MBS. Допуск fullauto относится к сходимости MBS.

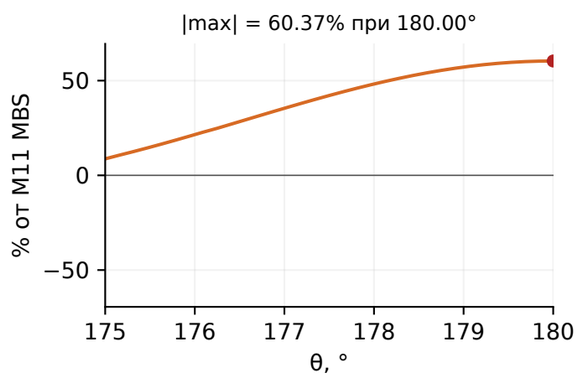


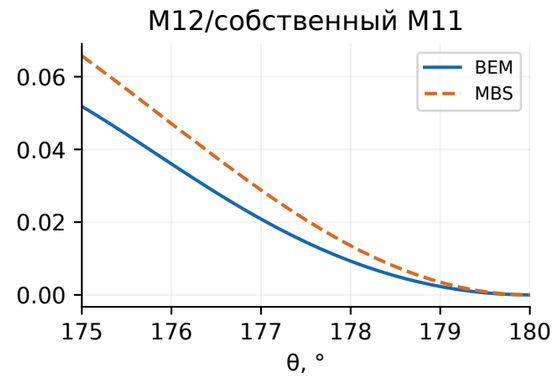
ka=20: ненулевые элементы и ошибки относительно MBS

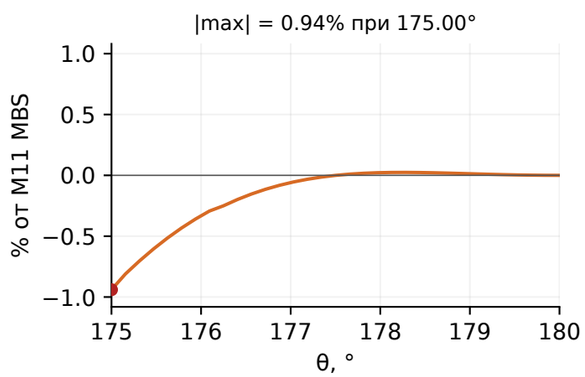
Кривые: M11 размерный; остальные Mij / собственный M11 каждого метода.

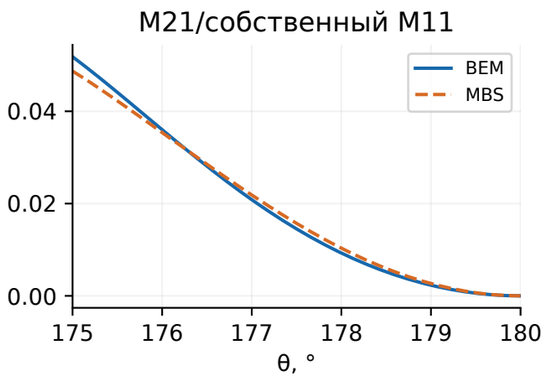
Ошибка исходных элементов: 100 (Mij BEM − Mij MBS) / M11 MBS.

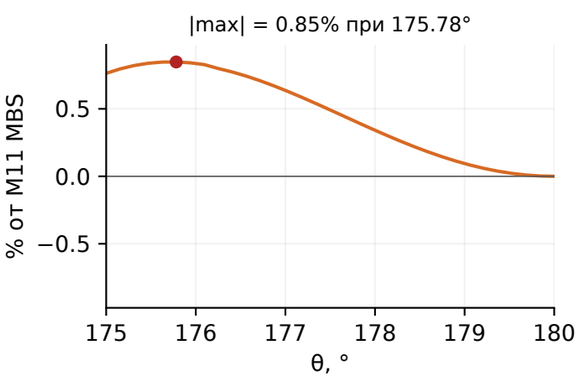


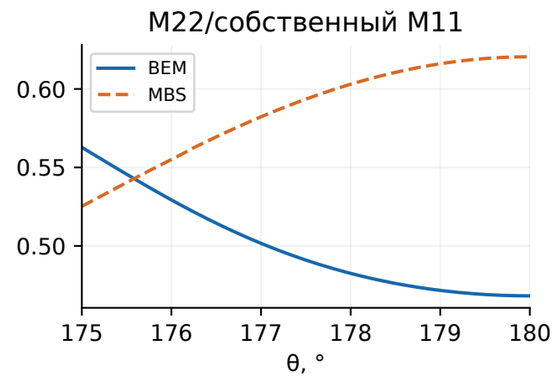


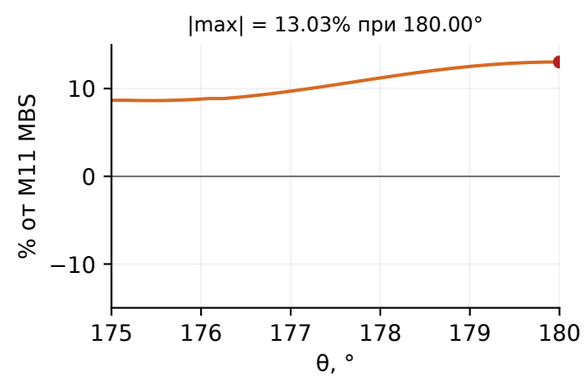


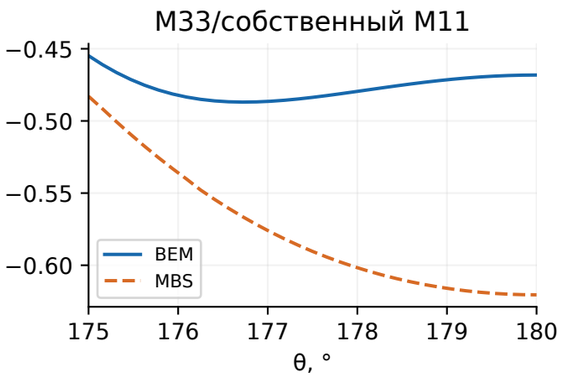


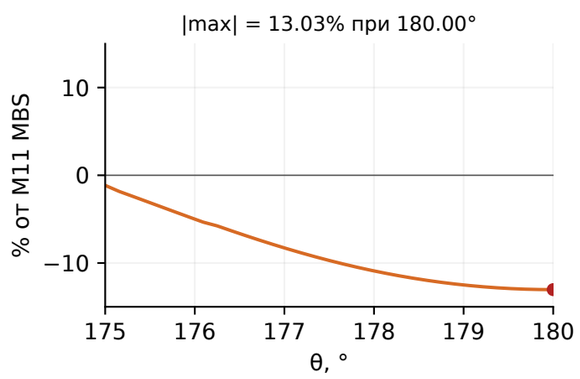


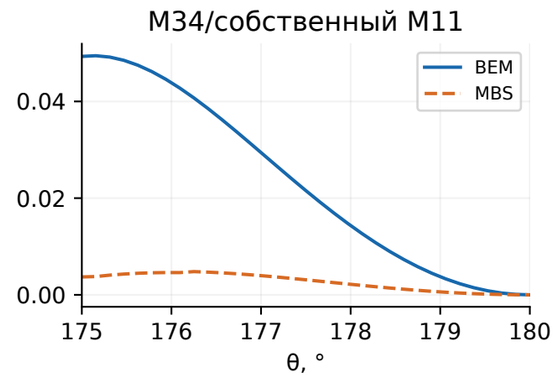


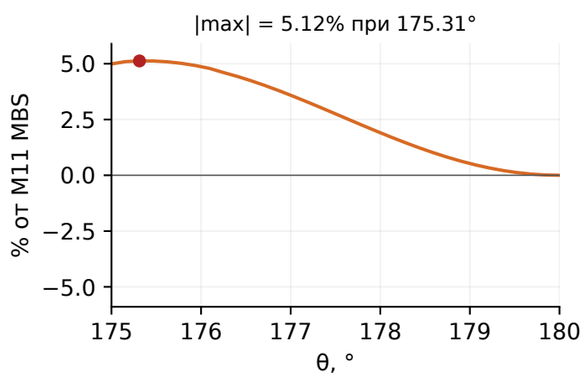


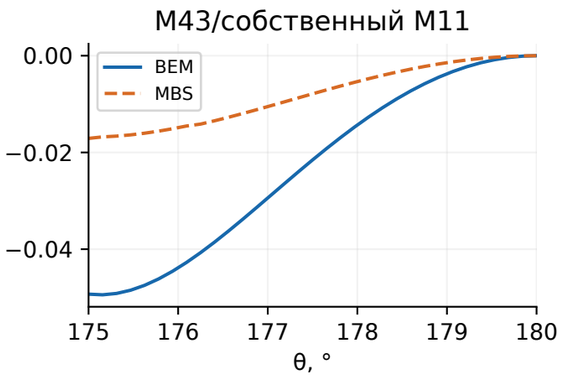


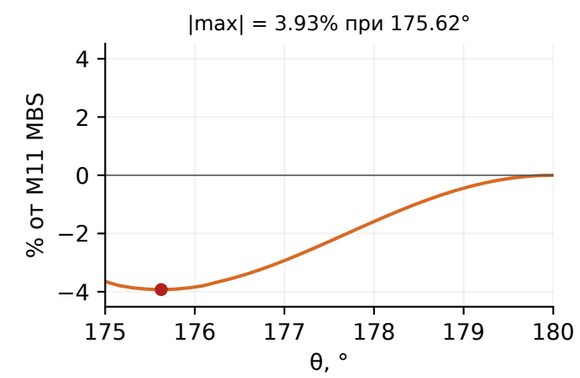


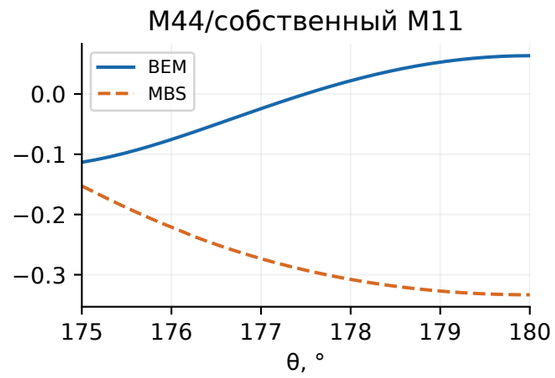


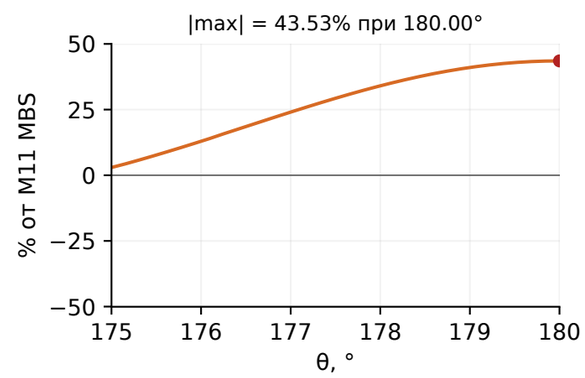








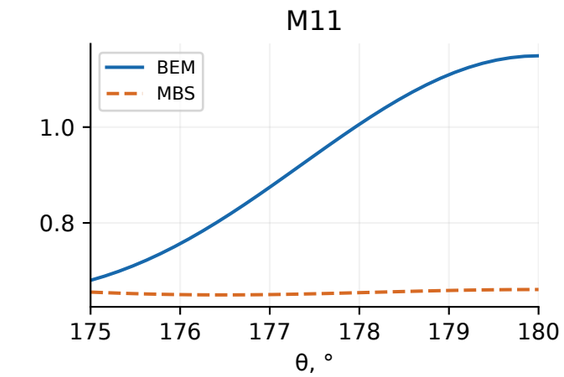


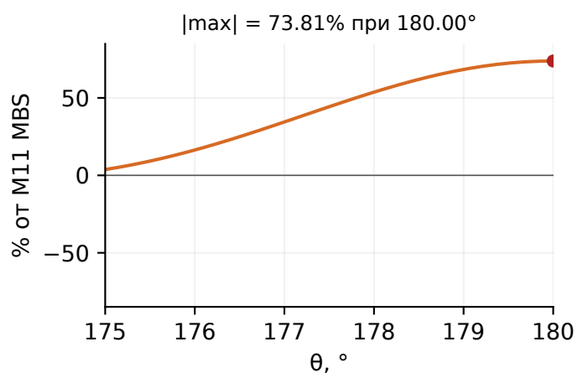


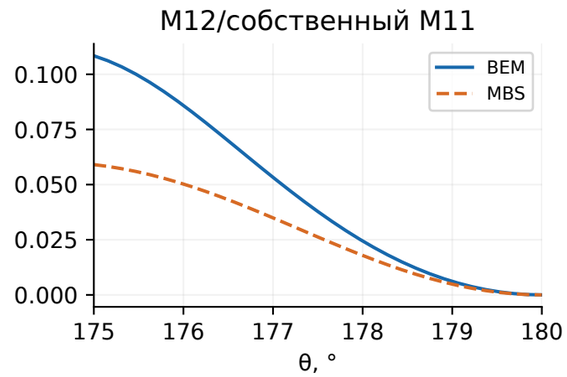
ka=25: ненулевые элементы и ошибки относительно MBS

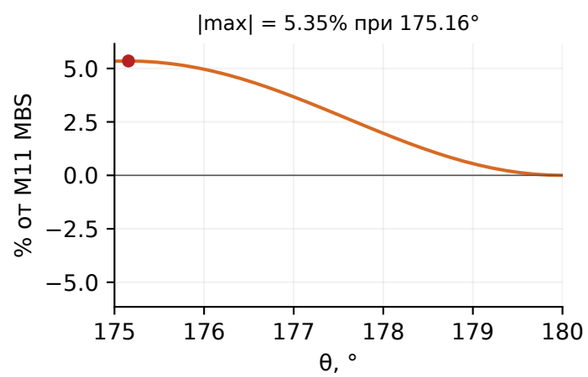
Кривые: M11 размерный; остальные Mij / собственный M11 каждого метода.

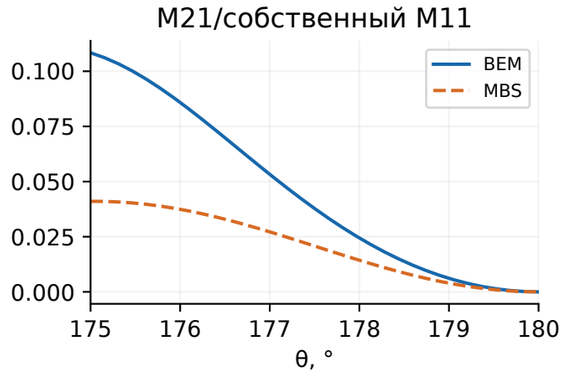
Ошибка исходных элементов: 100 (Mij BEM − Mij MBS) / M11 MBS.

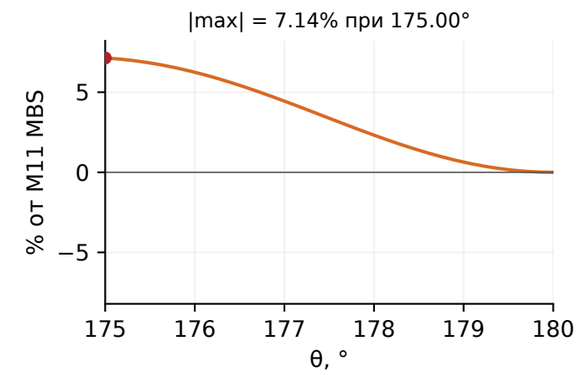


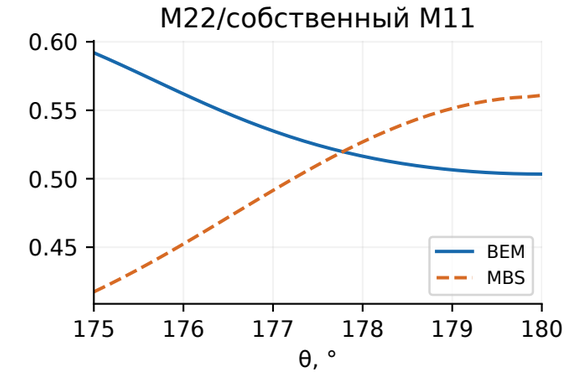


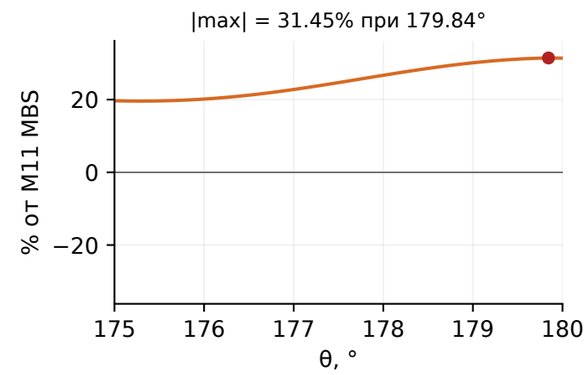


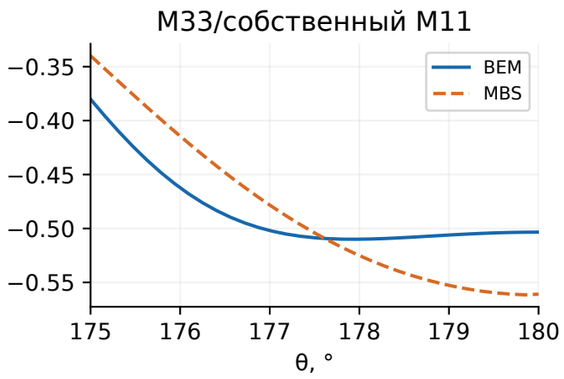


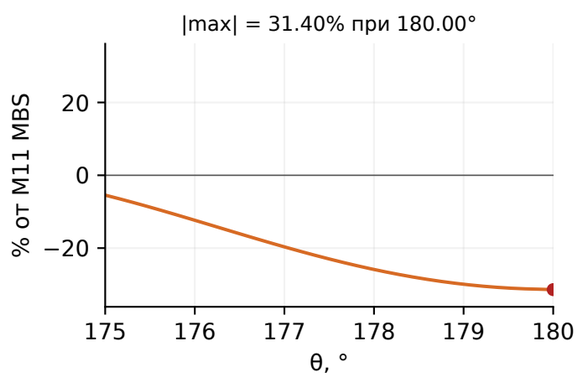


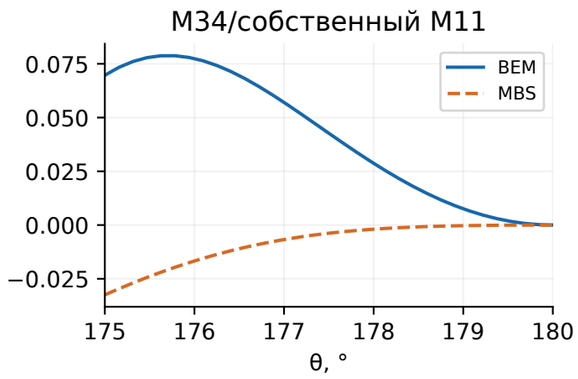


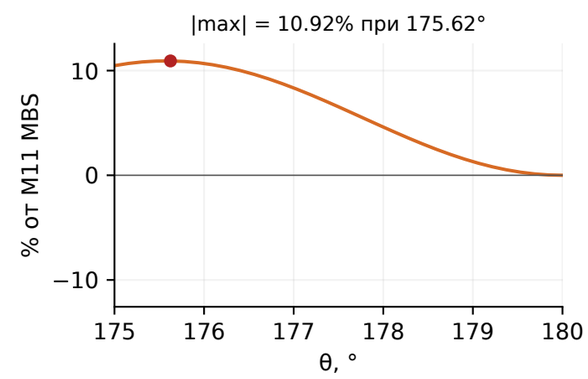


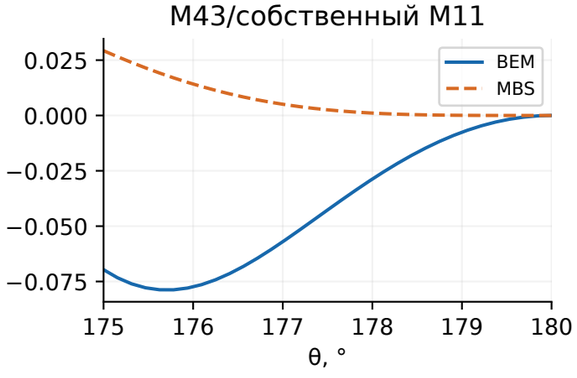


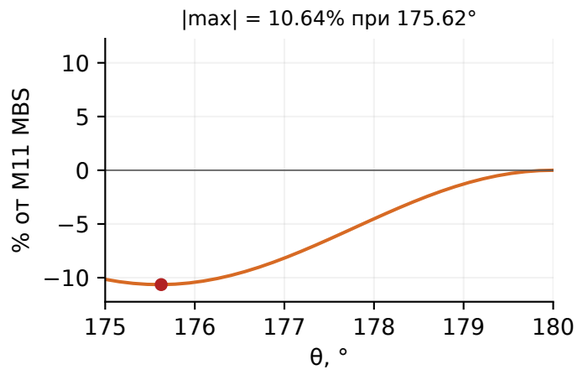


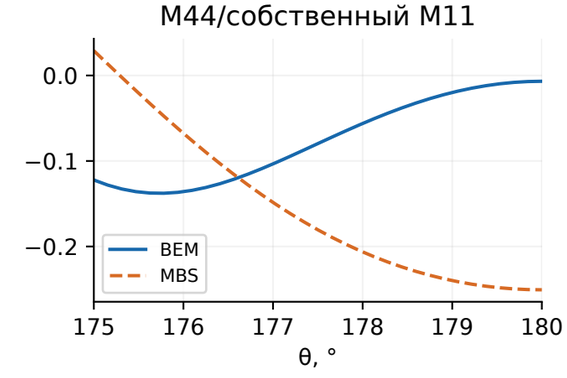


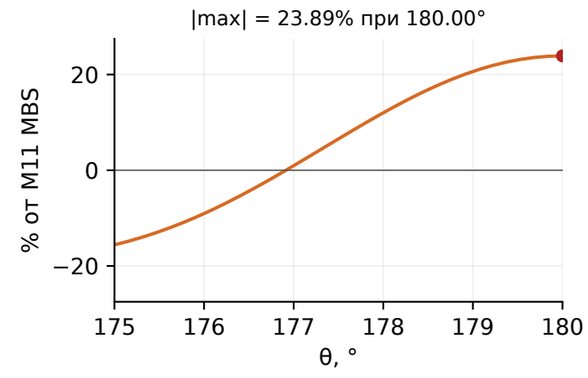








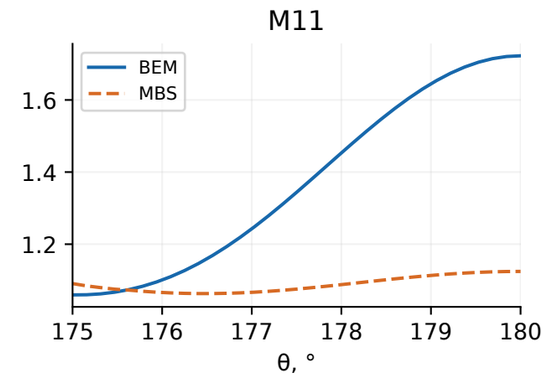


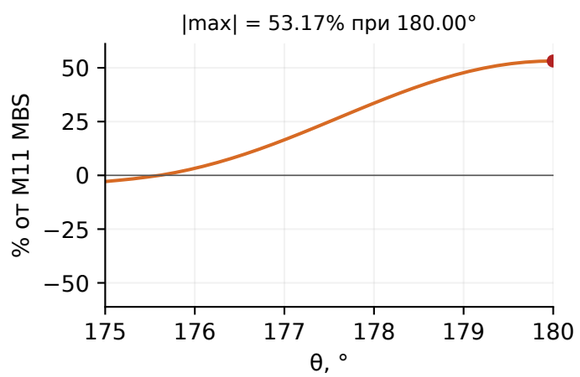


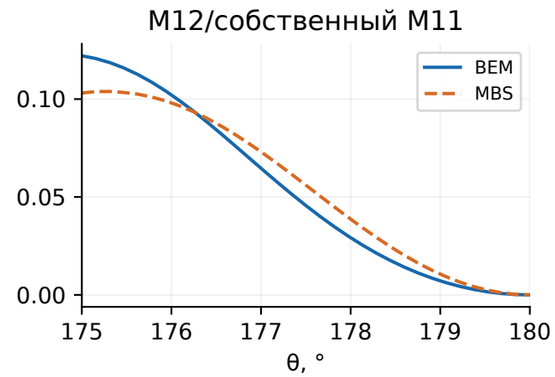
ka=30: ненулевые элементы и ошибки относительно MBS

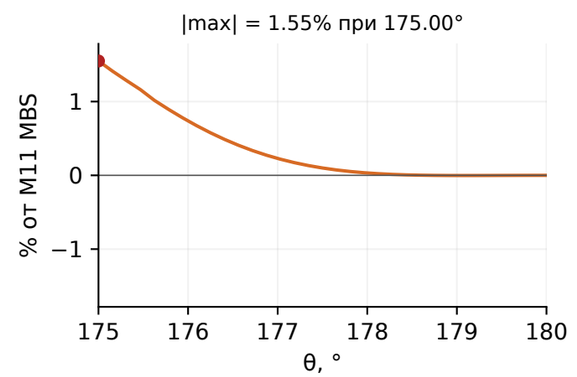
Кривые: M11 размерный; остальные Mij / собственный M11 каждого метода.

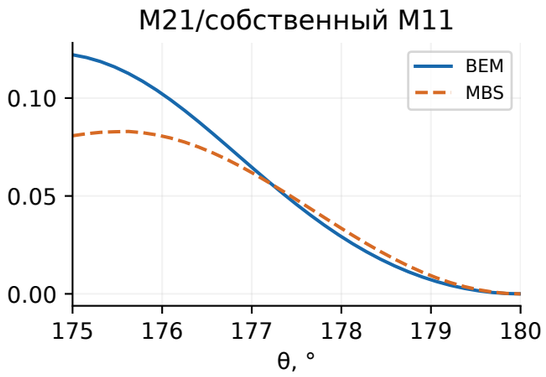
Ошибка исходных элементов: 100 (Mij BEM − Mij MBS) / M11 MBS.

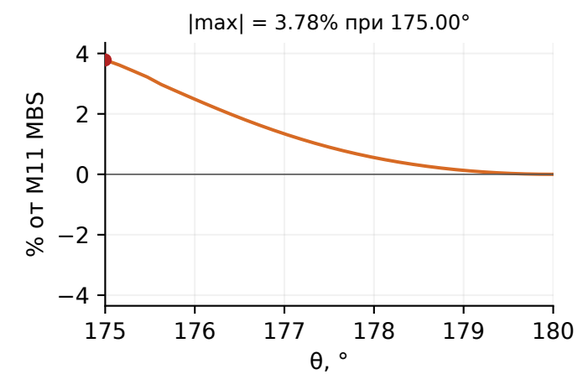


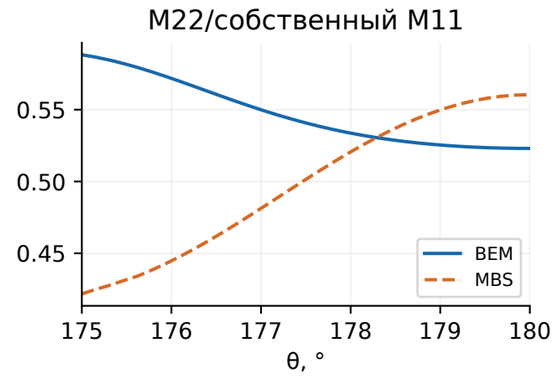


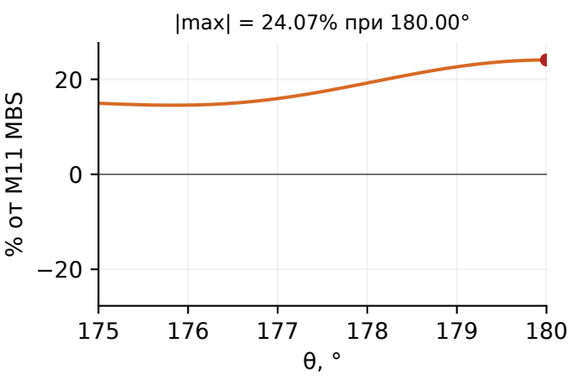


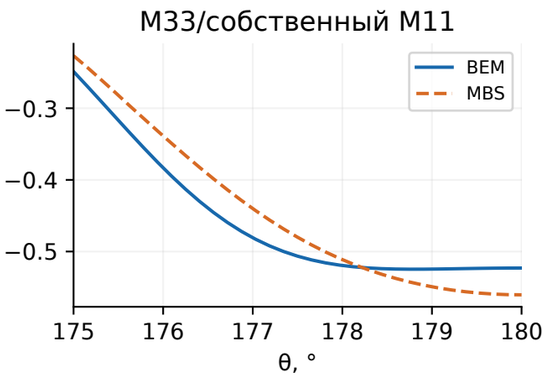


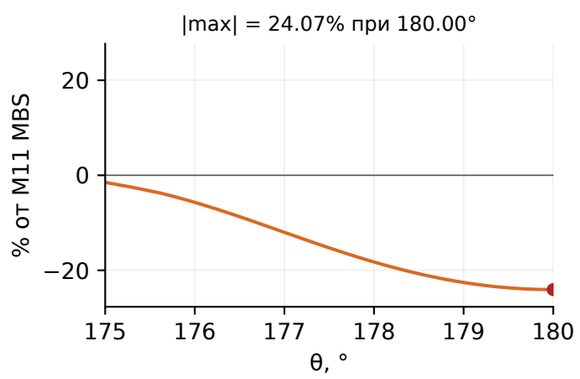


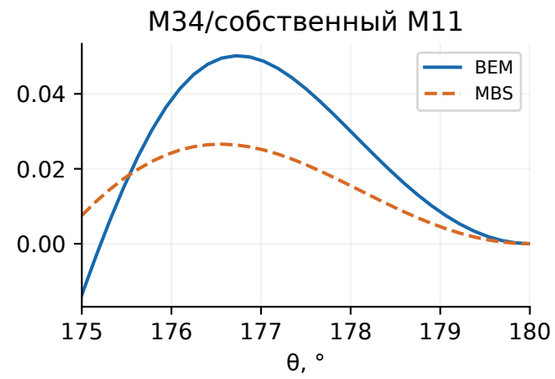


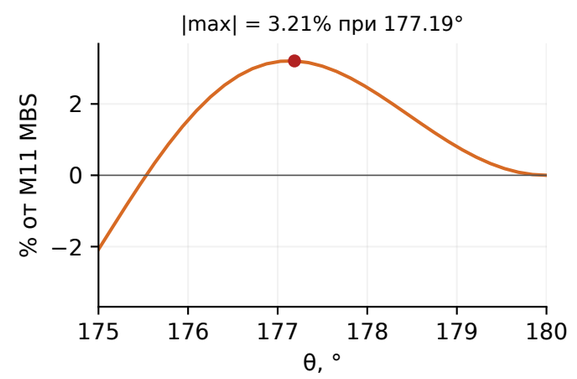


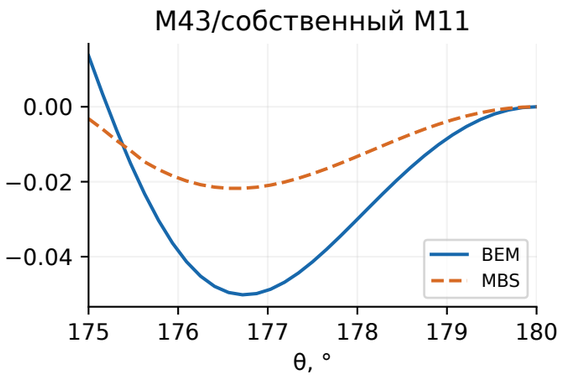


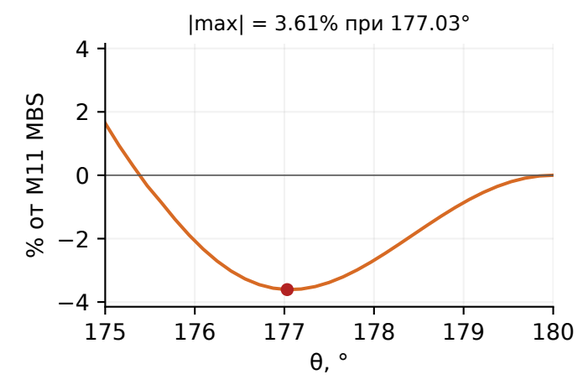


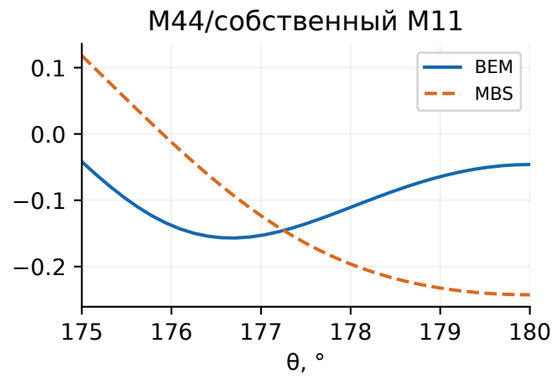


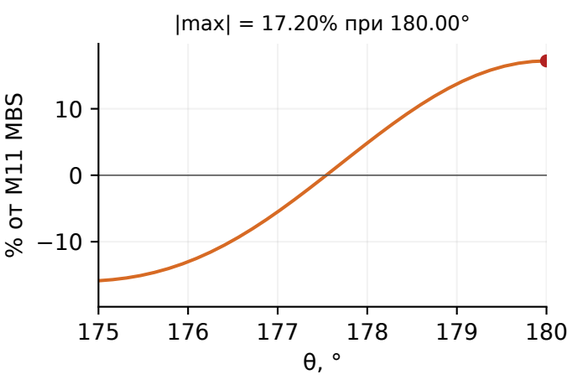








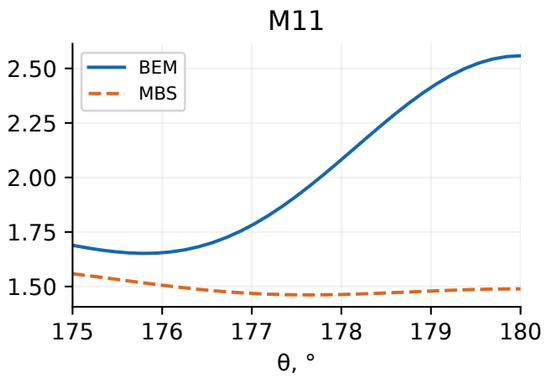


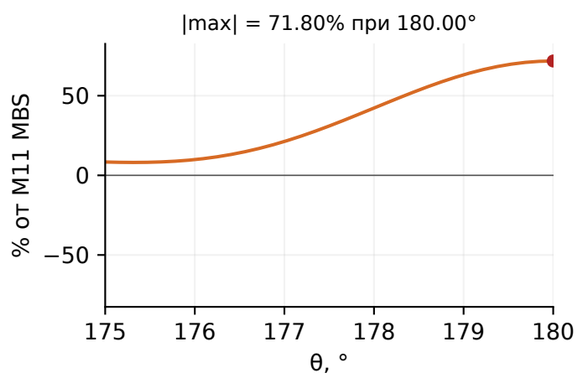


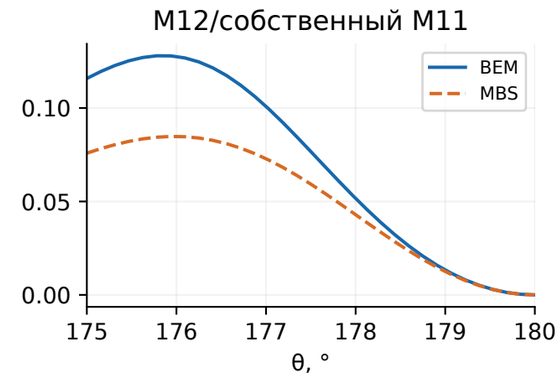
ka=35: ненулевые элементы и ошибки относительно MBS

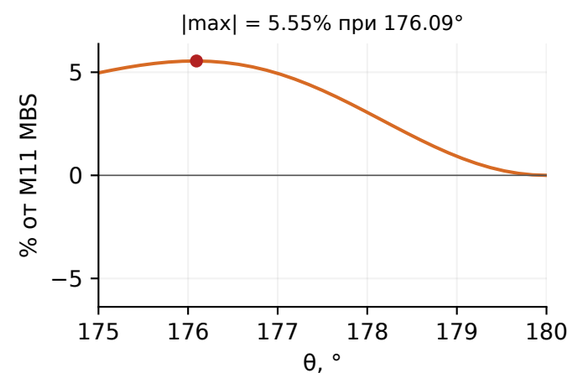
Кривые: M11 размерный; остальные Mij / собственный M11 каждого метода.

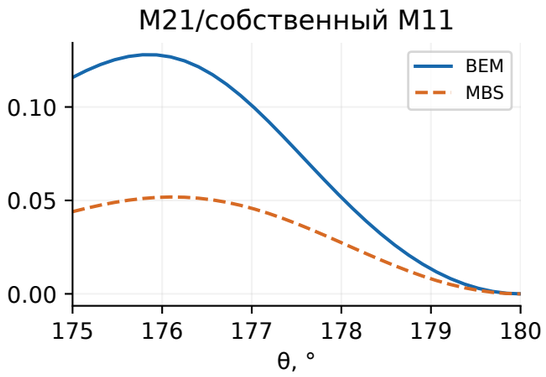
Ошибка исходных элементов: 100 (Mij BEM − Mij MBS) / M11 MBS.

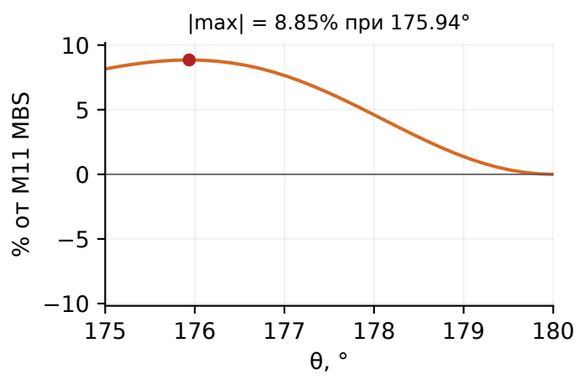


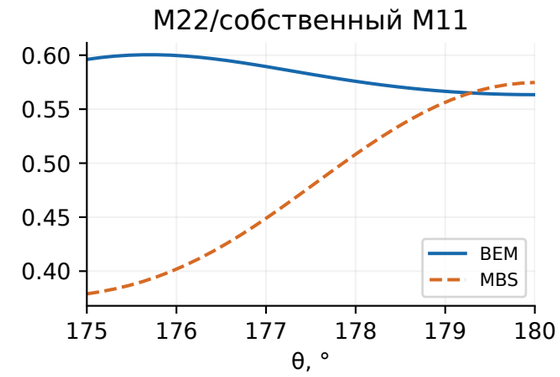


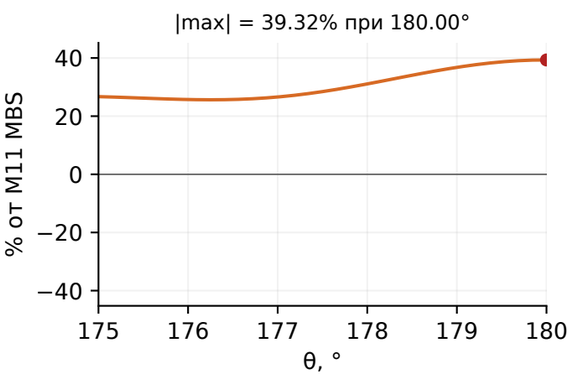


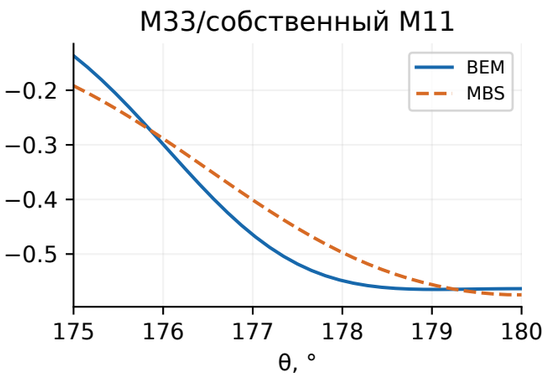


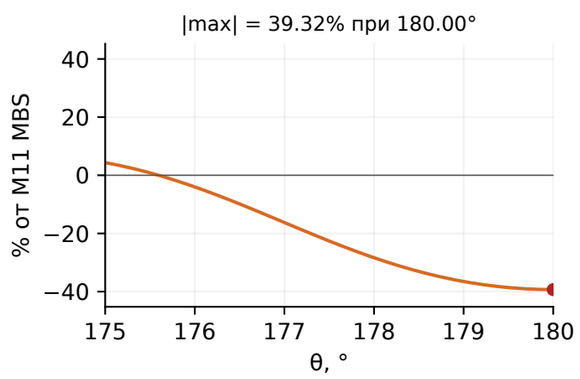


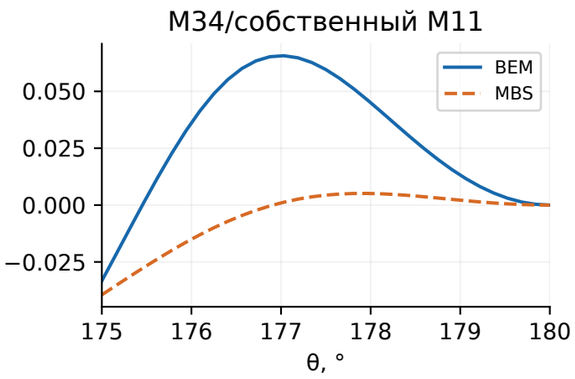


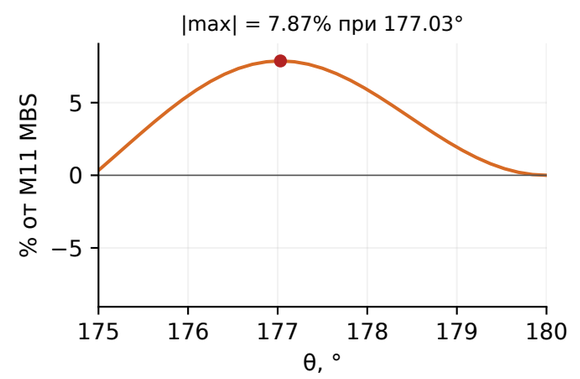


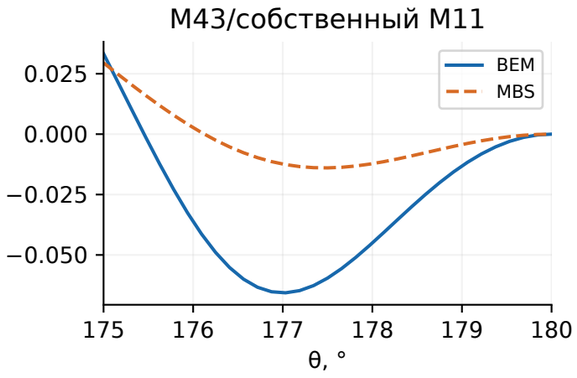


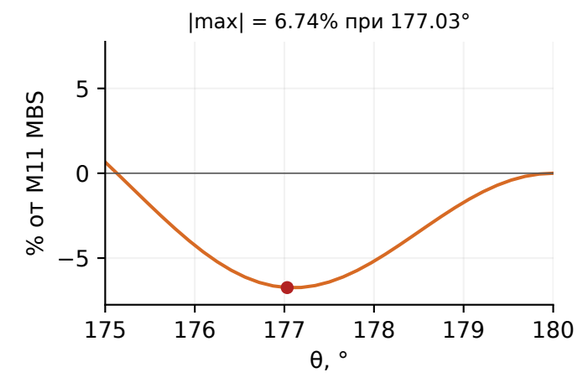


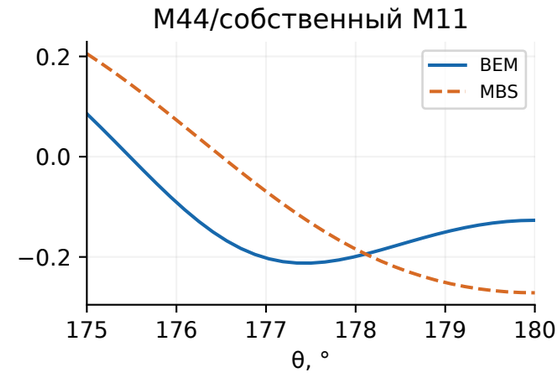


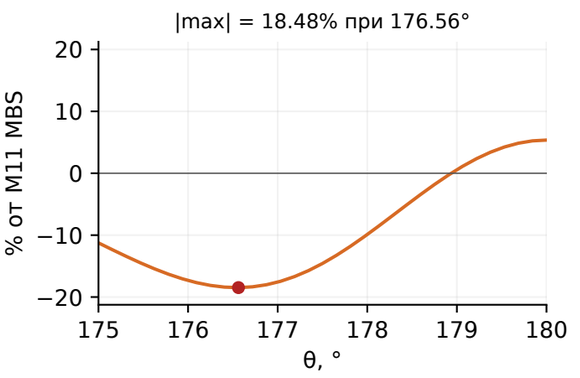








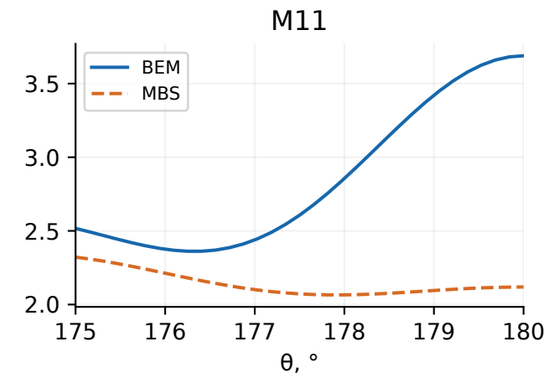


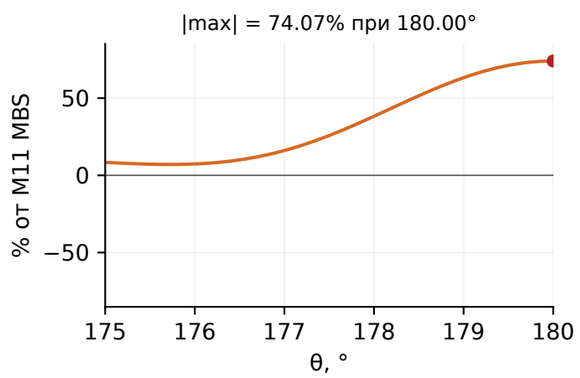


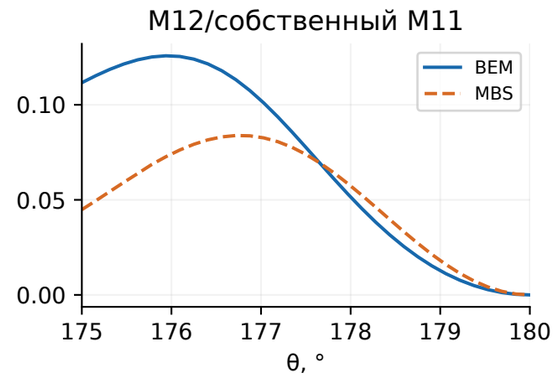
ka=40: ненулевые элементы и ошибки относительно MBS

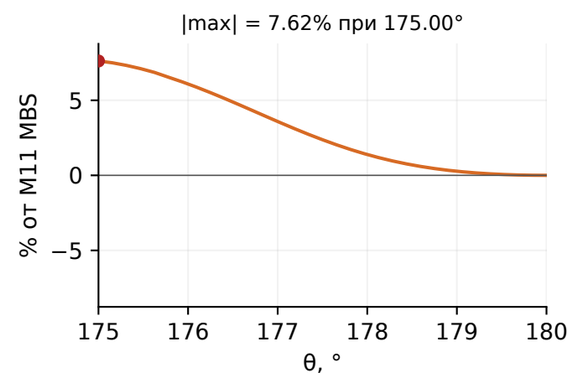
Кривые: M11 размерный; остальные Mij / собственный M11 каждого метода.

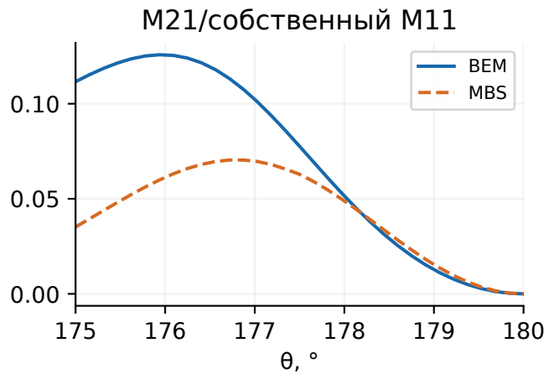
Ошибка исходных элементов: 100 (Mij BEM − Mij MBS) / M11 MBS.

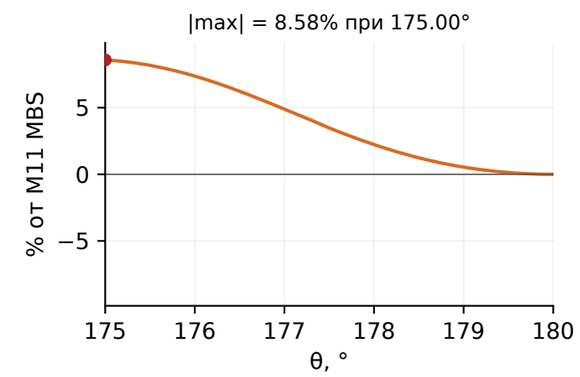


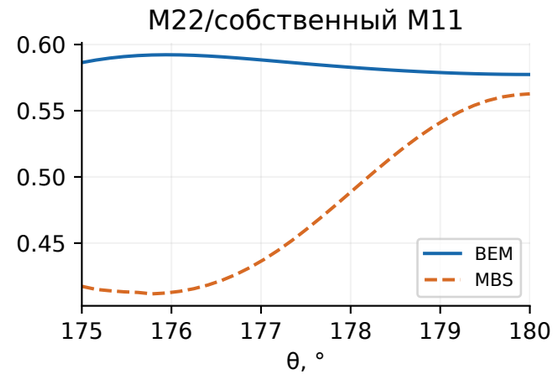


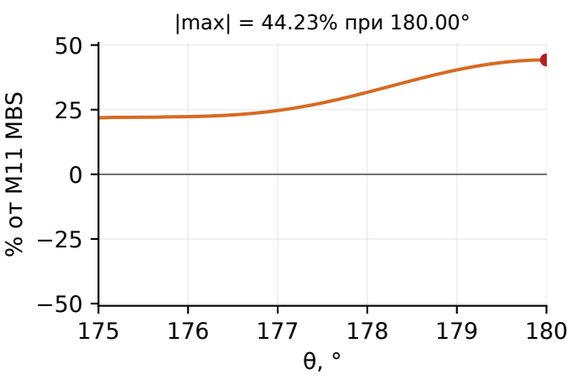


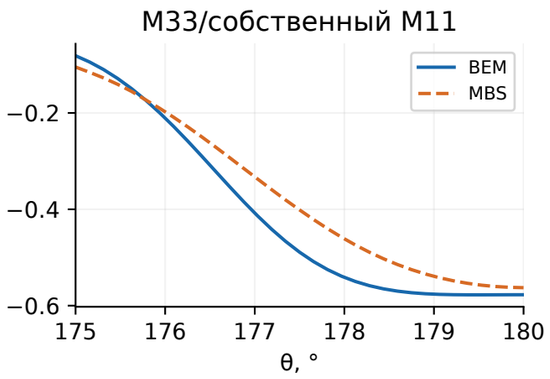


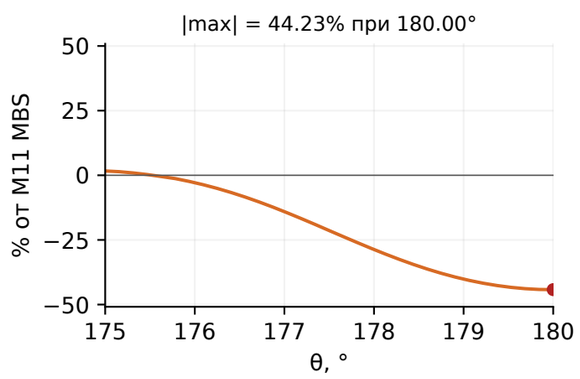


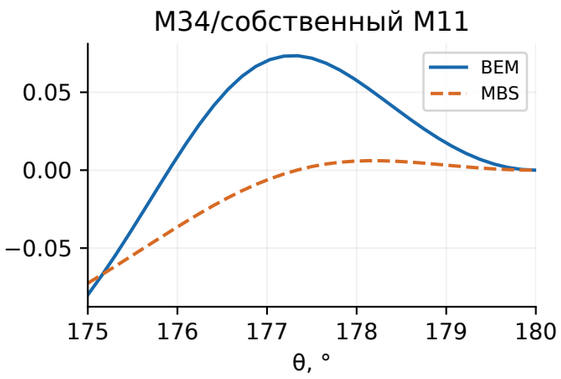


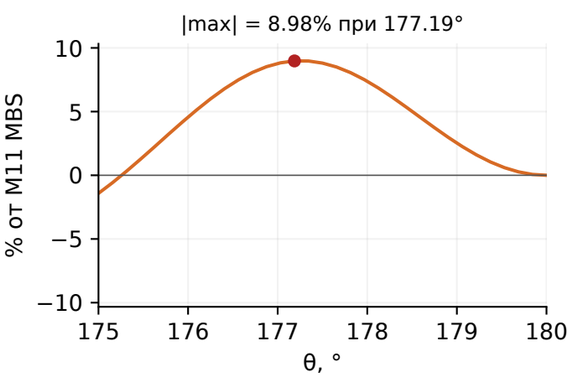


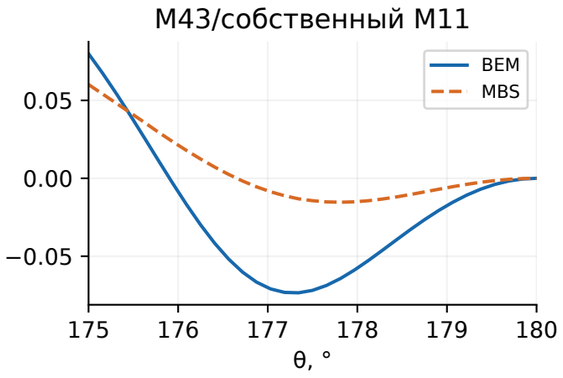


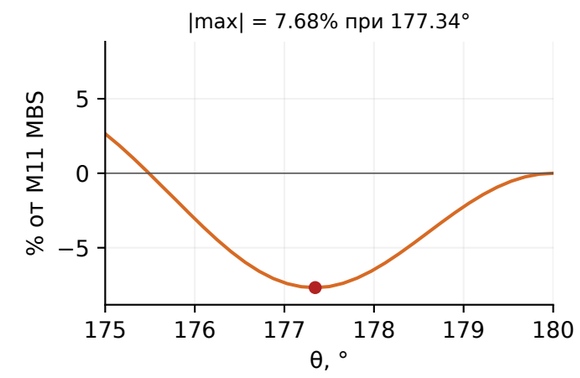


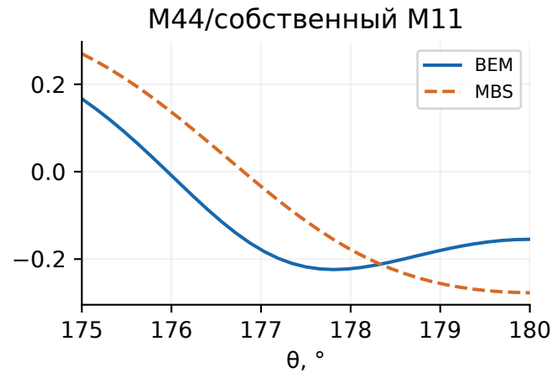


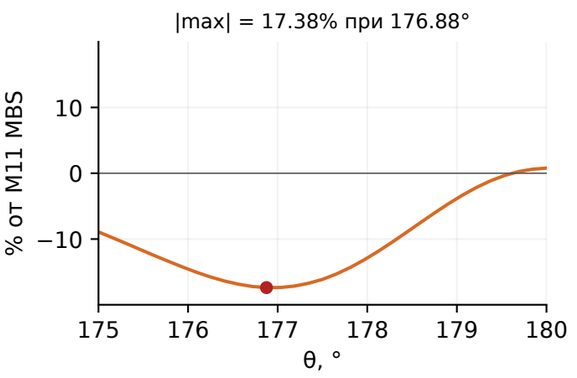












Интегральные ошибки относительно MBS — Исходные элементы

100 ∫|Q(BEM) − Q(MBS)| sinθ dθ / ∫|Q(MBS)| sinθ dθ; θ = 175…180°

Элемент	ka=20	ka=25	ka=30	ka=35	ka=40
M11	30.537	29.145	14.990	21.918	18.969
M12	7.578	99.163	7.476	66.640	71.990
M13	—	—	—	—	—
M14	—	—	—	—	—
M21	23.294	167.200	30.562	172.042	106.896
M22	17.035	47.417	34.858	63.590	58.391
M23	—	—	—	—	—
M24	—	—	—	—	—
M31	—	—	—	—	—
M32	—	—	—	—	—
M33	12.338	37.276	25.656	37.087	43.129
M34	1038.575	642.073	99.158	374.829	174.362
M41	—	—	—	—	—
M42	—	—	—	—	—
M43	256.544	738.675	141.844	349.117	183.509
M44	81.311	88.615	96.453	112.125	85.655

«—»: нулевой по симметрии элемент либо тождественное M11/M11.
Ошибка здесь — расхождение методов; 1% fullauto не означает 1% согласия с BEM.

Интегральные ошибки относительно MBS — Элементы / собственный M11

100 ∫|Q(BEM) − Q(MBS)| sinθ dθ / ∫|Q(MBS)| sinθ dθ; θ = 175…180°

Элемент	ka=20	ka=25	ka=30	ka=35	ka=40
M11	—	—	—	—	—
M12	24.124	67.266	10.892	44.174	54.657
M13	—	—	—	—	—
M14	—	—	—	—	—
M21	4.182	124.398	25.466	135.230	82.500
M22	11.018	17.345	19.651	34.286	32.905
M23	—	—	—	—	—
M24	—	—	—	—	—
M31	—	—	—	—	—
M32	—	—	—	—	—
M33	13.853	7.926	8.388	10.570	14.707
M34	824.355	546.851	76.415	323.819	157.213
M41	—	—	—	—	—
M42	—	—	—	—	—
M43	188.329	626.519	112.952	283.184	155.039
M44	82.594	89.783	99.666	98.893	76.538

«—»: нулевой по симметрии элемент либо тождественное M11/M11.
Ошибка здесь — расхождение методов; 1% fullauto не означает 1% согласия с BEM.